

Karun Joseph E

Final year B.Tech ECE Student

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Bangalore, India

Electronics and Communication Engineering student with hands-on experience in embedded hardware, PCB development, hardware bring-up, system integration, and embedded firmware. Experienced in sensor interfacing, SPI/UART communication, data acquisition, hardware validation, debugging, and production testing across aerospace, automotive/radar, and electronics manufacturing environments.

Experience

VIT Vellore (TI-Sponsored Program)

May 2026 - Present

Embedded Software Intern

- Developing embedded firmware on the TI J722S heterogeneous multi-core SoC for AWR2243 mmWave radar integration, using TI MCU+ SDK, FreeRTOS, SPI (MCSPI), and mmWaveLink for device initialization, configuration, and data acquisition
- Developed UART-based CLI utilities and debugged the hardware-software interface between the J722S and radar subsystem.

Team Sammar

April 2024 – Present

Electronics Engineer

- Designed and developed embedded avionics electronics and PCBs, integrating sensors, GPS, telemetry, payload control, RF communication, and flight-data acquisition systems.
- Worked with ESP32/STM32-based embedded systems, sensor interfaces, and communication protocols including SPI, I²C, UART, LoRa, and XBee/ZigBee.
- Supported hardware bring-up, embedded integration, troubleshooting, testing, and validation across flight electronics and payload subsystems.
- Built an ESP32-based solid-rocket propulsion testbench for load-cell data acquisition, integrating HX711, pressure sensing, SD-card logging, LoRa telemetry, and remote control.

Nokia Solutions & Networks

June 2025 – July 2025

SMT & Industrial Engineering Intern

- Designed a Raspberry Pi-based ESD workstation monitoring system with real-time dashboard integration for industrial safety and automation.
- Contributed to an ongoing automatic UUT quality-verification project.

Flex Electronics

May 2025 – June 2025

Production Intern

- Performed functional validation of Cisco switches and hubs; and developed a Python-based SFP fault-detection dashboard to support hardware troubleshooting and production diagnostics.

Education

Vellore Institute of Technology, Vellore

August 2023 - Present

B. Tech Electronics and Communications Engineering – **CGPA: 9.02**

Technical Skills

Hardware & Electronics: PCB Design, Prototyping, Sensor Integration, Hardware Bring-up, Data Acquisition, Telemetry, RF Communication, SMT/Mixed-Technology Soldering, Rework

Programming: C, Embedded C, C++, Python

Embedded: RTOS, STM32, ESP32, TI J722S (Cortex-R5F), Embedded Debugging, Hardware-Software Integration

Interfaces & Protocols: SPI (MCSPI, MIBSPI), UART, I²C, GPIO, MIPI CSI-2

Networking/RF: SFP, LoRa, XBee/ZigBee, BLE, RF Communication, mmWave Radar

Engineering Practices: Software Requirements, Verification & Validation, Unit Testing Fundamentals, Git Workflows

Tools: VS Code, KiCad, Altium Designer, EasyEDA, Bitbucket, Git.

Selected Projects

STM32 Flash Data Logger with Compression

- Developed a flash-aware STM32 data-logging architecture using DPCM prediction, Zigzag transformation, and Rice entropy coding to compress high-rate sensor data before SPI flash storage.
- Achieved 84.1% space savings (6.28:1 compression) while reducing flash page usage from 63 to 10 pages; 3rd Place, SanDisk Hackathon 2026.

CanSat Flight Computer PCB & Avionics

- Developed an STM32 based flight computer integrating GPS, sensors, XBee telemetry, payload control, and flight-data acquisition.
- Designed and integrated embedded electronics for reliable sensor acquisition, telemetry, and mission control.

Propulsion Testbench PCB for Solid Rocket Motors

- Developed an ESP32-based embedded data-acquisition system integrating HX711 load-cell measurement, pressure sensing, SD-card data logging, LoRa telemetry, and remote ignition trigger.
- Integrated multiple hardware peripherals and developed firmware for real-time measurement, logging, and communication.

ESD Compliance Monitoring System

- Raspberry Pi-based industrial ESD monitoring system with real-time dashboard integration.

Soft Skills: Problem Solving, Cross-Functional Collaboration, Technical Communication, Adaptability, Analytical Thinking